

Foundational Concepts of Analytics

1) Business Analytics

- The use of data, statistical methods and model to support business decision making
- Not just reporting (that's BI) and not purely technical data work (that's closer to data analytics). It is decision-focused.

2) Data Analytics

- The technical process of inspecting, cleaning, transforming and analysing data.
- Not always tied to business decisions. a data analyst might analyse data without making strategic recommendation

3) Business Intelligence (BI)

- Tools and system (like dashboards, reports) used to summarise past and current data.
- Not predictive or future focused. BI answers 'what happened' not 'what should we do next'

4) Data-driven decision making

- A management approach where decisions are based on data rather than intuition.
- ~~not~~ a tool or technique, it's a mindset or culture.

5) Big Data

- Extremely large and complex datasets defined by the 3 vs, Volume, Velocity, Variety.
- Not just 'a lot of data'. It specifically refers to data that traditional tools struggle to handle.

06) Descriptive Analytics

- Analysing historical data to understand what happened.
- It doesn't explain causes or predict the future.
- Example:** 'Sales increased by 10% last month.'

07) Diagnostic Analytics

- Investigating data to understand why something happened.
- It's not just reporting. It involves deeper analysis like correlations or root causes.
- Example:** 'Sales increased due to a marketing campaign.'

08) Predictive Analytics

- Using statistical models and machine learning to forecast what is likely to happen.
- Not guaranteed outcomes, only probabilities.
- Example:** 'Sales are likely to grow by 15% next quarter.'

09) Prescriptive Analytics

- Recommending actions based on predictions of what should we do.
- Not just predicting, it suggests decisions or strategies.
- Example:** Increase marketing spend to maximise sales growth.

10) Data Literacy :

- The ability to read, understand, analyse and communicate data.
- Not limited to analysts, everyone in a business should have it.

- Distinction

- Data Analytics vs Business Analytics

Data Analytics - working with data

Business Analytics - Using data for decisions

- BI vs Analytics

BI - past and present reports

Analytics - deeper insights + future thinking

- Descriptive - Diagnostic - predictive - prescriptive

What happened - why - what will happen - what to do.

Data, Measurement and Reporting

11) Structured Data

- Data organised in a fixed format, typically rows and columns (eg Excel tables, SQL databases)
- Not flexible or free-form like text or images.

Example: Customer table with Name, Age, purchase Amount

12) Unstructured Data

- Data that doesn't follow a predefined format, like text, images, audio, video,
- It not easily stored in traditional tables without processing

Example: Customer reviews, Social media posts

13) Data warehouse

A Centralised system that stores large amounts of structured data for reporting and analysis.

- It not used for daily transactions (that's operational databases)

14) Data Mining

The process of discovering patterns, trends or relationships in large data sets.

- not just querying or reporting - It involves deeper pattern discovery

Example: finding that customers who buy bread also buy milk

15) Data Visualisation

presenting data using charts, graphs and visuals to make insights easy to understand.

- not just making things look pretty, It must communicate meaning clearly, e.g Bar chart Showing monthly Sales

16) Dashboard

- An Interactive, real time visual display of key data and metrics.
 - Not a static report, it updates and allows quick monitoring
- Example: power BI dashboard tracking daily revenue

17) Metric

- Any quantifiable measure used to track performance.
 - Not necessarily important or actionable.
- Example: Number of website visitors.

18) Key performance Indicator (KPI)

- A Critical metric directly tied to business goals and decision making.
- Not just any number, it must influence decisions and success.

Example: Conversion rate (if the goal is to increase sales)

Metric vs KPI

- All KPI are metrics but not all metrics are KPIs.

Example

- website visits
 - for a blog - just a metric
 - for an ad based business - a KPI (drives revenue)

Another.

Number of employees

- for most companies - metric
- for a company trying to reduce costs - KPI

19) Benchmark

- A standard or reference point used to compare performance.
 - Not your actual performance, it's what you compare against.
- Example: Industry average Conversion rate = 5%

20) Segmentation

- Dividing customers into groups based on shared characteristics.
 - not random grouping, it must be meaningful and useful.
- Example: Segment customers by age, location or buying behavior.

Distinction Summary

- Structured vs Unstructured data
 - Structured - organised (table)
 - Unstructured - messy (text, images)
- Data Visualisation vs Dashboard
 - Visualisation - single chart
 - Dashboard - collection of visuals (often live)
- Metric vs KPI
 - metric - any number
 - KPI - a number that matters for decisions

Customer and Sales KPIs

21) Conversion Rate

- The percentage of users who complete a desired action (eg purchase, sign-up)
- not total sales, it's about efficiency not volume.

Example: 1000 visitors, 50 purchases - Conversion Rate 5%.

22) Customer acquisition Cost (CAC)

- The average cost of acquiring one new customer (marketing + sales costs).

• not revenue, it's a cost metric, not profit.

Example: Spend R10 000 on ads - get 100 customers - CAC R100

23) Customer lifetime Value (CLV/LTV)

- The total profit a business expects to earn from a customer over their entire relationship.

• not just one purchase, it's long term value.

Example: A customer spends R500 per year for 3 years
CLV = R1500 (before cost)

24) Churn Rate

- The percentage of customers who stop doing business with you over a period.

• not about gaining customers only losing them.

example: 100 customers - 10 leave - Churn Rate = 10%

25) Customer Retention Rate

- The percentage of Customers who stay with your business
- Not separate from Churn, they are inversely related.

26) Net promoter Score (NPS)

- A measure of Customer loyalty based on the question:
'How likely are you to recommend us?'
- not a direct revenue metric, it measures Customer Sentiment.

Scale: 9-10 = promoters, 7-8 = passives, 0-6 = Detractors.

27) Average order value (AOV)

- The average amount spent per transaction.
 - Not total revenue, it's per order, not overall.
- example: 120 000 revenue from 100 orders - $AOV = R100$

28) Sales funnel

- The journey Customers take from awareness - Interest - decision - purchase
 - not a single metric - It's a process / stage model.
- example: website visit - add to cart - checkout - purchase.

The most Important Ratio: CLV: CAC

- CLV - what you earn from a Customer.
 - CAC - what you spend to get them
- It tells you if your business is profitable and sustainable.

- CLV: CAC = Value per Customer = Cost to acquire Customer.

How to Interpret it

- 1:1 - losing money (you spend as much as you earn)
- <1:1 - Unsustainable (you lose money per customer)
- 3:1 - Healthy (ideal benchmark in many industries)
- >5:1 - Could mean under-investing in growth.

Example

- $CAC = \$100$

- $CLV = \$600$

$$CLC : CAC = 6:1$$

- you earn 6x what you spend - very strong business model

Marketing and Campaign Analytics

29) Marketing Campaign

- A coordinated set of marketing activities (ads, emails, social media) designed to achieve a specific goal.
- Not a single ad or post, it's a planned, multi-channel effort.

example: A product launch campaign across Instagram, email and Google ads.

30) Post-Campaign Analysis

- The process of evaluating a campaign's performance after it ends.
- Not real-time tracking, it happens after execution to assess success. It uses: ROI, ROAS, CTR, incremental revenue.

31) Incremental Revenue

- The additional revenue generated directly because of the campaign.
- Not total revenue, only the extra amount above normal performance.

Example:

Normal Sales = R250 000

Campaign Sales = R280 000

Incremental Revenue = R30 000

32) Return On Investment (ROI)

• A measure of profitability: how much profit you made compared to cost.

• not just revenue, it considers costs and profit.

$$ROI = (\text{profit} \div \text{cost})$$

33. Return on ad spend (ROAS)

• Revenue generated for every rand spent on advertising

• not profit, it ignores costs beyond ad spend.

example:

Spend R1000 - earn R4000 ROAS = 4x

ROI vs ROAS

• ROI - measures profitability (after costs)

• ROAS - measures ad efficiency (revenue only)

34) Click-Through Rate (CTR)

• The percentage of people who click on an ad after seeing it.

• not conversions, it only measures clicks, not outcomes.

example: 1000 views, 50 clicks - CTR = 5%

35) Bounce Rate

• The percentage of visitors who leave a website without interacting.

• not about clicks, it measures engagement after landing

36) A/B Testing

- Comparing two versions (A vs B) to see which performs better

- Not guessing, It's data driven experimentation

example

ad A vs Ad B - measure CTR, Conversions, etc.

37) Lead Generation

- The process of Capturing potential Customers' information (eg emails)

- Not actual sales, These are prospects, not buyers yet.

example: sign up form for a free ebook.

38) Marketing Attribution

- Determining which marketing channel (eg email, social media, search ads) deserves credit for a conversion.

- Not always simple, Customers often interact with multiple channels.

example:

Customer sees Instagram ad - Clicks google ad - Buys - attribution decides who gets credit

How these Concepts Connect

1) Campaign flow

- You run a marketing campaign

- It generates - clicks (CTR)

 - Leads

 - website behavior (bounce rate)

2) Optimisation During Campaign

- Use A/B testing to improve:
 - Ads (increase CTR)
 - Landing pages (reduce bounce rate)

3) Measuring Success after Campaign

- Do post-campaign analysis using:
 - Incremental Revenue - what extra did we gain.
 - ROAS - did ads generate strong revenue.
 - ROI - was it actually profitable.

4) Understanding Impact

- Use Marketing Attribution to answer.